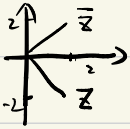
P51 Section 4

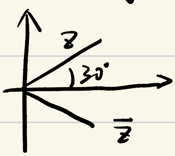
10. $2-2i$



x	y	r	θ
2	-2	$2\sqrt{2}$	$-\frac{\pi}{4}$

$2\left(\cos\left(-\frac{\pi}{4}\right)+i\sin\left(-\frac{\pi}{4}\right)\right) \Rightarrow 2e^{i\left(-\frac{\pi}{4}\right)}$

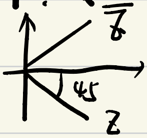
11. $2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)$



x	y	r	θ
$\sqrt{3}$	1	2	$\frac{\pi}{6}$

$2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right) \Rightarrow 2e^{i\frac{\pi}{6}}$

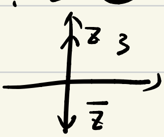
17. $\sqrt{2}e^{-i\frac{\pi}{4}}$



x	y	r	θ
1	-1	$\sqrt{2}$	$-\frac{\pi}{4}$

$\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right)+i\sin\left(-\frac{\pi}{4}\right)\right) \Rightarrow \sqrt{2}e^{-i\frac{\pi}{4}}$

18. $3e^{i\frac{\pi}{2}}$

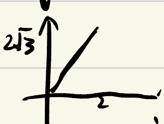


x	y	r	θ
0	3	3	$\frac{\pi}{2}$

$3\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right) \Rightarrow 3e^{i\frac{\pi}{2}}$

P52 Section 5

5 $(i+\sqrt{3})^2$



x	y	r	θ
2	$2\sqrt{3}$	4	$\frac{\pi}{3}$

$\Rightarrow 2+i\cdot 2\sqrt{3}$

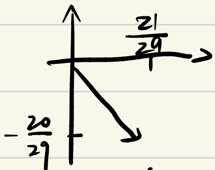
6. $\left(\frac{1+i}{1-i}\right)^2$

x	y	r	θ
-1	0	1	π

$\Rightarrow e^{\pi i} = -1$

15. $\dfrac{5-2i}{5+2i}$

x	y	r	θ
$\frac{21}{29}$	$-\frac{20}{29}$	1	-43.6°



$\dfrac{5-2i}{5+2i} = \dfrac{21}{29} - \dfrac{20}{29}i$

16. $\dfrac{1}{0.5(\cos 40^\circ + i\sin 40^\circ)}$

x	y	r	θ
1.53	-1.29	2	-40°

$\Rightarrow 2e^{i(-\frac{2}{9}\pi)}$

45. $(x+iy)^2 = (x-iy)^2 \Rightarrow$ ① $x=0, \; y \in R$
② $x \in R, \; y=0$

50. $|x+iy| = y-ix \Rightarrow x=0, \; y \geq 0$

59 $|z+3i| = 4$. Circle, center $(0,-3)$, $r=4$

62. $|z+1| + |z-1| = 8$

Ellipse, foci at $(1,0)$ $(-1,0)$; semi-major axis $x=4$

Section 12

6. $\cos 2z = \cos^2 z - \sin^2 z$

$$\frac{e^{iz\cdot 2}+e^{iz-2}}{2} \Rightarrow \frac{e^{i2z}+2+e^{-i2z}}{4} + \frac{e^{i2z}-2+e^{-i2z}}{4}$$

$$= \frac{e^{i2z}+e^{-i2z}}{2} \quad 成立$$

10. $\frac{d}{dz}\cosh z = \sinh z$

$$\frac{d}{dz}\frac{e^{z}+e^{-z}}{2} = \frac{e^{z}-e^{-z}}{2} = \sinh z \quad 成立$$

17. $\tanh iz = i\tan z$

$$\frac{\sinh z}{\cosh z} = \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}} \Downarrow \quad 成立$$

$$\tan z = \frac{e^{iz}-e^{-iz}}{i(e^{iz}+e^{-iz})} \Rightarrow \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$$

33. $\tan i = \frac{e^2-1}{e^2+1} i$

34. $\sin i\frac{\pi}{2} = \frac{e^{-\frac{\pi}{2}} - e^{\frac{\pi}{2}}}{2i} = \frac{e^{\frac{\pi}{2}} - e^{-\frac{\pi}{2}}}{2} i$

Section 14

14. $(2i)^{1+i} = e^{(i+1)\ln 2i}$ $\qquad 2i = 2e^{i\frac{\pi}{2}}$

$\ln 2i = \ln 2 + i\cdot\frac{\pi}{2}$

$(i+1)\ln(2i) = \ln 2 - \frac{\pi}{2} + (\ln 2 + \frac{\pi}{2})i$

$e^{(i+1)\ln 2i} = 2e^{-\frac{\pi}{2}}(i\cos(\ln 2) - \sin(\ln 2))$

17. $(i-1)^{i+1} = e^{(i+1)\ln(i-1)}$ $\qquad i-1 = \sqrt{2}e^{i\frac{3}{4}\pi}$

$\ln i-1 = \ln\sqrt{2} + i\frac{3}{4}\pi$

$(i+1)(\frac{1}{2}\ln 2 + i\frac{3}{4}\pi) = (\frac{3}{4}\pi + \frac{1}{2}\ln 2)i + \frac{1}{2}\ln 2 - \frac{3}{4}\pi$

$\Rightarrow \sqrt{2}\, e^{-\frac{3}{4}\pi} e^{i(\ln\sqrt{2} + \frac{3}{4}\pi)}$

Section 16.

11. ① $2\sin\theta\cos\theta + \cos 3\theta \cdot 2\sin\theta + \cdots = \sin 2n\theta$

$$\frac{e^{2i\theta}-e^{-2i\theta}}{2i} + \cdots \quad (e^{i3\theta}+e^{-i3\theta})(e^{i\theta}-e^{-i\theta})$$ 互消

$$\Rightarrow \sim = \frac{e^{i2n\theta}-e^{-i2n\theta}}{2i} = \sin 2n\theta$$

② $\sin\theta + \sin 3\theta + \cdots + \sin(2n-1)\theta = \frac{\sin^2 n\theta}{\sin\theta}$

$$\sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$$

$$\sin\theta\sin(2n-1)\theta = -\frac{1}{4}\left(e^{i2n\theta}+e^{-i2n\theta}-e^{-i(2n-2)\theta}-e^{i(2n-2)\theta}\right.$$

$$\Rightarrow \frac{e^{i2\theta}+e^{-i2\theta}-2}{-4} + \bigcirc - \cancel{\bigcirc}$$

$$= \frac{e^{i2n\theta}+e^{-i2n\theta}-2}{-4} = \sin^2 n\theta = \frac{e^{i2n\theta}+e^{-i2n\theta}-2}{-4}$$

$\Rightarrow$ 成立

Section 17

25 a) $\overline{\cos z} = \cos \overline{z}$

$$\cos z = \frac{\cos x(e^{-y}+e^{y}) + i\sin x(e^{-y}-e^{y})}{2}$$

$$\overline{\cos z} = \frac{\cos x(e^{-y}+e^{y})^{2} + i\sin x(e^{y}-e^{-y})}{2}$$

$$\cos \overline{z} = \frac{e^{i\overline{z}}+e^{-i\overline{z}}}{2} = \frac{(\cos x + i\sin x)e^{y} + (\cos x - i\sin x)e^{-y}}{2}$$

$\Rightarrow \overline{\cos z} = \cos \overline{z}$

b) $\overline{\sin z} = \sin \overline{z}$?

$$\sin z = \frac{e^{ix-y}-e^{-ix+y}}{2i} = \frac{\cos x(e^{-y}-e^{y})i - \sin x(e^{-y}+e^{y})}{-2}$$

$$\overline{\sin z} = \frac{-\sin x(e^{-y}+e^{y}) - \cos x(e^{-y}-e^{y})i}{-2}$$

$$\sin \overline{z} = \frac{-\sin x(e^{y}+e^{-y}) - \cos x(e^{-y}-e^{y})i}{-2}$$

$\Rightarrow \sin \overline{z} = \overline{\sin z}$ Yes